

Logically Equivalence, Predicate and Quantifiers

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Based on the lecture note uploaded on SPeCTRUM

Sem I, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is logically equivalence and determine if two statements are logically equivalence with truth table.
- ☐ Understand the algebraic laws of logic and apply it to determine if two statements are logically equivalence.
- ☐ Apply the algebraic laws of logic to simplify a statement.
- ☐ Understand what is predicate and its domain and determine the truth value of a statement from a predicate.
- ☐ Understand what is a universal and existential quantifier and determine the truth value of a statement involving them.
- ☐ Negate the statement involving quantifier.

Logical Equivalence

Two propositions are said to be ^{logically equivalent} ~~logical equivalence~~ if their truth value match for all possible component propositions value, i.e, the corresponding columns in the truth table are identical.

We use $P \equiv Q$ to represent two propositions P and Q are (logically) equivalent.

Note: $P \equiv Q$ if and only if $P \leftrightarrow Q$ is a tautology.

Example: $\sim p \wedge \sim q$ VS $\sim (p \vee q)$ VS $\sim (p \wedge q)$

p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$p \vee q$	$\sim (p \vee q)$	$p \wedge q$	$\sim (p \wedge q)$
T	T	F	F	F	T	F	T	F
T	F	F	T	T	T	F	F	T
F	T	T	F	T	T	F	F	T
F	F	T	T	T	F	T	F	T

↑ ↑

$\sim p \vee \sim q \equiv \sim(p \wedge q)$ because their truth values agree for all possible combinations of the truth value of p and q .

$\sim p \vee \sim q$ and $\sim(p \vee q)$ are not logically equivalent because $\sim p \vee \sim q$ is true but $\sim(p \vee q)$ is false when p is true and q is false.

Logical Equivalence

Example: Show that $p \rightarrow q \equiv \sim p \vee q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	$\sim p$	q	$\sim p \vee q$
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T

Example: Show that the negation of $p \rightarrow q$ is $p \wedge \sim q$.

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

p	q	$\sim q$	$p \wedge \sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

Logical Equivalence

Example: Show that the converse of $p \rightarrow q$ is not equivalent to $p \rightarrow q$. (logically)

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Example: Show that the contrapositive of $p \rightarrow q$ is equivalent to $p \rightarrow q$.

p	q	$p \rightarrow q$	$\sim p$	$\sim q$	$\sim q \rightarrow \sim p$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

Algebraic laws of logic

For propositions p, q, r and any tautology t and contradiction c , we have the following **algebraic laws of logic**:

Commutative

Associative

Distributive

Identity

Negation

Double negation

Idempotent

* De Morgan's

Universal bound

Absorption

Negation of t and c

Conditional equiv

$$p \vee q \equiv q \vee p$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$p \vee (q \wedge r) \equiv$$
$$(p \vee q) \wedge (p \vee r)$$

$$p \vee c \equiv p$$

$$p \vee \sim p \equiv t$$

$$\sim \sim p \equiv p$$

$$p \vee p \equiv p$$

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

$$p \vee t \equiv t$$

$$p \vee (p \wedge q) \equiv p$$

$$\sim t \equiv c$$

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \wedge q \equiv q \wedge p$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$p \wedge (q \vee r) \equiv$$

$$(p \wedge q) \vee (p \wedge r)$$

$$p \wedge t \equiv p$$

$$p \wedge \sim p \equiv c$$

$$p \wedge p \equiv p$$

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$p \wedge c \equiv c$$

$$p \wedge (p \vee q) \equiv p$$

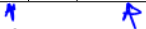
$$\sim c \equiv t$$

Algebraic laws of logic

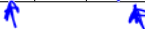
Exercise: Try to use the truth table to show the algebraic laws of logic.

Identity Law:

$p \vee c \equiv p$		
p	c	$p \vee c$
T	F	T
F	F	F



$p \wedge t \equiv p$		
p	t	$p \wedge t$
T	T	T
F	T	F



Absorption Law:

$p \vee (p \wedge q) \equiv p$			
p	q	$p \wedge q$	$p \vee (p \wedge q)$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F



$p \wedge (p \vee q) \equiv p$			
p	q	$p \vee q$	$p \wedge (p \vee q)$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	F	F



Algebraic laws of logic

De Morgan's Law:

$\sim (p \vee q) \equiv \sim p \wedge \sim q$						
p	q	$p \vee q$	$\sim (p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Converse:

$\sim (p \wedge q) \equiv \sim p \vee \sim q$						
p	q	$p \wedge q$	$\sim (p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

Ex: (John ^p is smart) but (lazy) ^q $\sim(p \wedge q)$
 $\equiv \sim p \vee \sim q$

Negation: John is not smart
 or John is not lazy.

$$\sqrt{2} \leq 2 < \pi \rightarrow \boxed{\sqrt{2} \leq 2} \text{ and } \underline{2 < \pi}$$

$\underbrace{\hspace{1.5cm}}_p \quad \wedge \quad \underbrace{\hspace{1.5cm}}_q$
 $\text{or } \underbrace{\hspace{1.5cm}}_{q_r}$

Negation: $\sqrt{2} > 2$ or $2 \geq \pi$

$$\sim p \vee \sim q$$

$$\sim(r \vee s) \equiv \sim r \wedge \sim s$$

$$\begin{array}{ccc} r & \vee & s \\ \uparrow & & \uparrow \\ \sqrt{2} < 2 & & \sqrt{2} = 2 \end{array}$$

Algebraic laws of logic

Example: Show that the negation of $p \rightarrow q$ is $p \wedge \sim q$.

$$\begin{aligned}\sim(p \rightarrow q) &\equiv \sim(\sim p \vee q) && \text{Conditional equivalence.} \\ &\equiv \sim(\sim p) \wedge \sim q && \text{De Morgan's Law.} \\ &\equiv p \wedge \sim q && \text{Double Negation Law.}\end{aligned}$$

Example: Simplify $(p \vee \sim q) \wedge \sim(p \wedge q)$.

$$\begin{aligned}(p \vee \sim q) \wedge \sim(p \wedge q) &\equiv (p \vee \sim q) \wedge (\sim p \vee \sim q) && \text{De Morgan's Law.} \\ &\equiv (p \wedge \sim p) \vee (\sim q) && \text{Distributive Law.} \\ &\equiv \text{false} \vee (\sim q) && \text{Negation Law.} \\ &\equiv \sim q && \text{Identity Law.}\end{aligned}$$

Algebraic laws of logic

Example: Simplify $\sim (p \vee q) \vee (\sim p \wedge q)$.

Exercise -

Predicate

A sentence is a **predicate** (or **propositional function**) if

- it contains a finite number of **predicate variables** $x_1, \dots, x_n$, and
- it is not a statement, but it becomes a statement when particular value are substituted for the variables.

The **universe of discourse** or **domain** D of a predicate variable is the set of all values that may be substituted for the variable.

Example:

Let $P(x) : x \text{ is an animal}$ and domain = {apple, cat, computer, elephant, horse}.

$P(\text{apple}) : \text{apple is an animal. is a false statement.}$

$P(\text{cat}) : \text{cat is an animal is a true statement.}$

$P(\text{computer})$ is a false statement.

$P(\text{elephant})$ and $P(\text{horse})$ are true statements.

Predicate

Example:

Let $P(x, y) : x \geq y$ and

domain = $\{(x, y) \mid x = 1, 2, \text{ or } 3, \text{ and } y = 3 \text{ or } 4\}$.

$P(1, 3) : 1 \geq 3$ is false.

$P(1, 3)$ $P(3, 3)$ $P(1, 4)$ $P(2, 4)$ $P(3, 4)$
F T F F F

Let $P(x) : x \text{ is prime} \rightarrow x \text{ is odd}$. What is the truth value of each of the following statements?

$P(17)$
T

$P(2)$
F

$P(3)$
T

$P(0)$
T

Let $P(x, y) : x^2 - y^2 = 0 \leftrightarrow x = y$. What is the truth value of each of the following statements?

$P(1, 1)$
T

$P(1, -1)$
F

$P(1, 0)$
T

P	Q	$P \leftrightarrow Q$
T	T	T
F	T	F
T	F	F
F	F	T

Universal quantifier

A **universal statement** has the form “For all $x \in D$, $P(x)$ is true”. Here, D is called the **domain** and $P(x)$ is some properties related to x , known as **predicate**.

The **universal quantifier** is denoted by $\forall$, and can be read as “for all”, “for each”, “for every” or “for any”.

A universal statement is true if and only if $P(x)$ is true for all x in the domain

A universal statement is false if and only if $P(x)$ is false for at least one x in the domain. Such value of x is called a **counterexample** to the universal statement.

Universal quantifier

Example:

Determine the truth value of the following universal statement. If the universal statement is false, suggest a counterexample to the universal statement.

- ① $\forall x \in \{-1, 0, 3, 4, 5\}, 2x$ is an even integer. *A true statement.*

$$2(-1) = -2 \text{ even} \quad 2(3) = 6 \checkmark$$

$$2(0) = 0 \text{ even} \quad 2(4) = 8 \checkmark$$

$$2(5) = 10 \checkmark$$

- ② $\forall x \in \{x \mid 0 \leq x \leq 1\}, x < 5x.$

$0 < 0$
false.

A false statement. Counterexample: $x = 0$

- ③ $\forall x \in \mathbb{Z}, x + 1$ is nonnegative. *A false statement.*

*↑
set of all
integers.*

Counterexample: $x = -100$

Existential quantifier

A **existential statement** has the form “There exists $x \in D$, such that $P(x)$ is true”.

The **existential quantifier** is denoted by $\exists$, and can be read as “there exists”, “for some”, “for at least one” or “there is some”.

An existential statement is true if and only if $P(x)$ is true for at least one x in the domain.

An existential statement is false if and only if $P(x)$ is false for all x in the domain.

Unique Existence $\exists!$ is used to denote “there exists one and only one”.
 $\exists!x, P(x)$ is read as “There exists a unique x such that $P(x)$ ”.

Existential quantifier

Example:

Determine the truth value of the following existential statement.

① $\exists x \in \mathbb{R}, x \leq x + 1.$

↓
set of all real numbers.

True. E.g. $x=0$ $0 \leq 0+1$

② $\exists x \in \mathbb{Z}, x^2 = 2.$

False. 2 is not a perfect square.

③ $\exists! x \in \mathbb{R}, 2x + 3 = 5x + 7.$

True.

$$3x = -4$$

$$x = \frac{-4}{3}$$

is the only answer.

Negating statement with one quantifier

To negate a statement with one quantifier, we do the following:

- Change the quantifier ($\forall$ to $\exists$ or vice versa)
- Negate the predicate expression

$$\sim (\forall x, P(x)) \equiv \exists x, \sim P(x)$$

$$\sim (\exists x, P(x)) \equiv \forall x, \sim P(x)$$

Example:

Quantified statement	Negation
$\forall x \in \mathbb{R}, x^2 > 0$	$\exists x \in \mathbb{R}, x^2 \leq 0$
All roses are red.	Some rose is not red.
Some student failed.	All students passed.

Negating Statement with one quantifier

Translate each of the following statements into symbolic form using quantifier. Then write their negations in logical expression form and in English.

$P(x): x \text{ is prime}$ $Q(x): x \text{ is odd.}$ $D: \mathbb{R}.$

Every prime number is odd. (Domain: All real numbers)

Symbolic form: $\forall x \in \mathbb{R}, P(x) \rightarrow Q(x)$

Negation: $\exists x \in \mathbb{R}, P(x) \wedge \sim Q(x)$

There is a real number such that it is prime but it is even.
Some prime number is even.

No triangle has four sides. (Domain: All polygons) $D: \text{The set of all polygons.}$

Symbolic form: $\forall x \in D, P(x) \rightarrow \sim Q(x)$

$P(x): x \text{ is a triangle.}$

$Q(x): x \text{ has four sides.}$

Negation: $\exists x \in D, P(x) \wedge Q(x)$

There is a triangle that has four sides.

Negating Statement with one quantifier

Translate each of the following statements into symbolic form using quantifier. Then write their negations in logical expression form and in English.

$P(x)$: x eats cheeseburger $Q(x)$: x eats cheesecake.

Some Malaysian eats cheeseburgers but does not eat cheesecake.

(Domain: All Malaysians) D : the set of all Malaysians.

Symbolic form: $\exists x \in D, P(x) \wedge \sim Q(x)$

$$\sim(P \wedge Q)$$

Negation: $\forall x \in D, \sim P(x) \vee Q(x)$

$$\sim p \vee q$$

All Malaysians do not eat cheeseburgers or do eat cheesecake.

Statements with multiple quantifiers

It is possible for a statement to contain more than one quantifiers.

Example:

Translate each of the following statements into symbolic form using quantifiers. Which of the following statements is true?

- ① Given any positive number, there is another positive number that is smaller than the given number.

Symbolic form: $\forall x \in \mathbb{R}^+, [\exists y \in \mathbb{R}^+, y < x]$

Eg: $x=1$, we can take $y=0.5$

$x=0.0001$, we can take $y=0.00001$

A true statement.

- ② There is a real number x such that for every real number y , the sum of the x and y is zero.

Symbolic form: $\exists x \in \mathbb{R}, [\forall y \in \mathbb{R}, x+y=0]$

For $x \neq 0$, Take $y=0$, $x+y \neq 0$

For $x=0$, take $y=1$

A false statement.

Negation: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x+y \neq 0$. True.

Negation: $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+, x > y$ false

$x > y$ false

$x=2$ x
 $x=1$ x

$x=0$ x
 $x=1$ x
 $x=2$ x

$$\forall y \in \mathbb{R}, [\exists x \in \mathbb{R}, x+y=0]$$

$$\begin{aligned} y &= -2 \\ y &= -1 \text{ take } x=1 \checkmark \\ \rightarrow y &= 0 \text{ take } x=0 \checkmark \\ y &= 1 \text{ take } x=-1 \checkmark \end{aligned}$$

Here, for any real number y , we can always take $\underline{x = -y}$ and $x+y=0$. So this is a true statement.

$$\text{eg: } \forall x \in \mathbb{R}, [\exists y \in \mathbb{R}, (\forall z \in \mathbb{R}, x+y+z=0)]$$

Negating Statement with more than one quantifier

When negating a statement with multiple quantifiers, we simply negate each quantifying component in turn.

Example:

$$\sim (\forall x, \exists y, P(x, y) \text{ is true}).$$



$$\exists x, \sim [\exists y, P(x, y) \text{ is true}].$$



$$\exists x, \forall y, \sim (P(x, y) \text{ is true})$$



$$\exists x, \forall y, P(x, y) \text{ is false.}$$